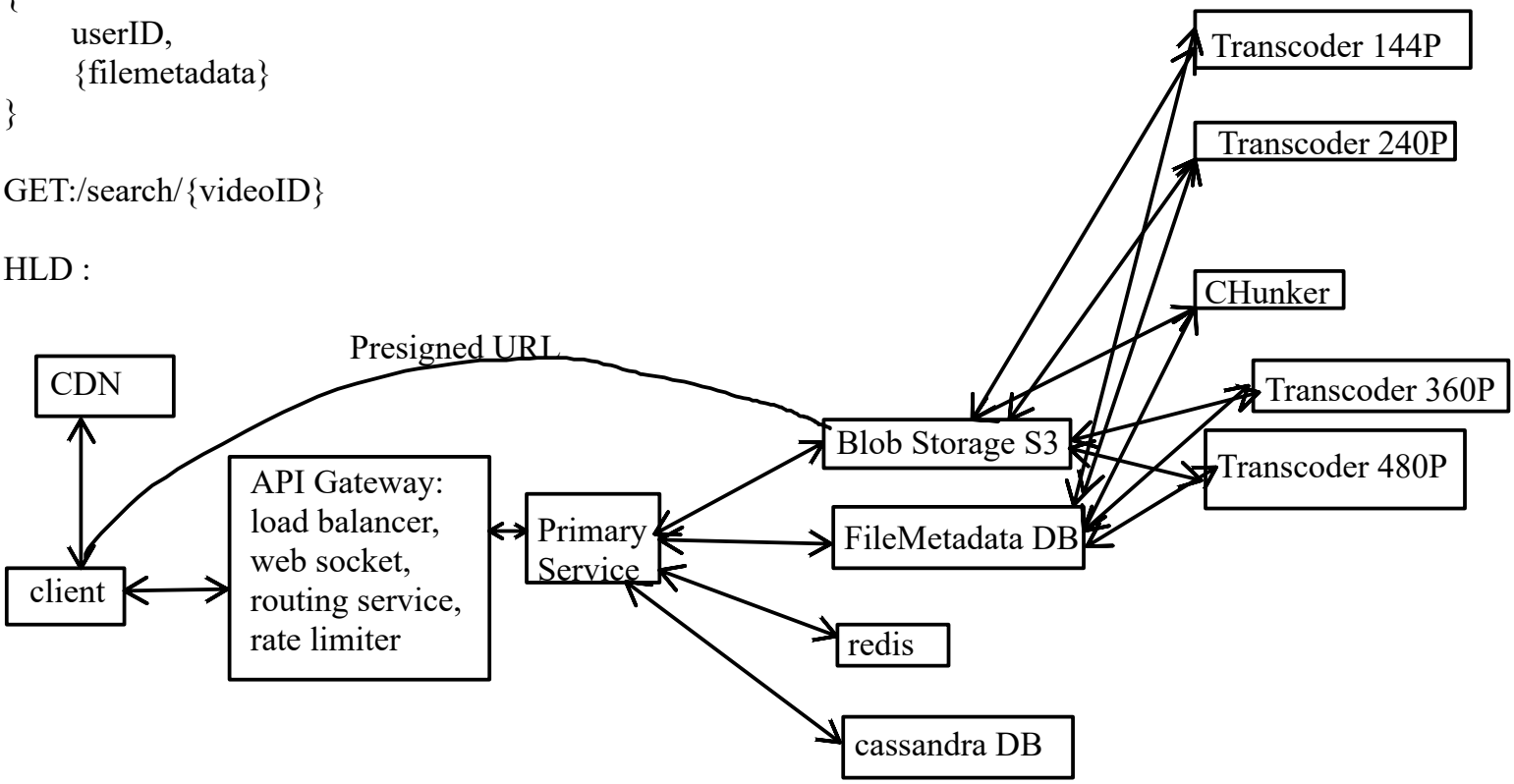
Youtube:

Functional requirements:
user should be able to open/download any video based on his preference,
user should be able to search for any videos based on his preference,
video quality should get adjusted based on his preference,

Non-functional requirements:
Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (here eventual consistency is fine , but here we need 999.99% uptime i.e Availability>>Consistency)

Core Entities:
user,
video,
channel,
feed

API :
POST:/upload/video
{
userID,
{filemetadata}
}

GET:/search/{videoID}

HLD :